Supplementary Material: Can Aesthetic Emotions Explain Individual Taste? Investigating an Interpretable Route to Personalized Image Aesthetics

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This document gives the experimental details needed to audit the main paper, reports the extended model and mediator analyses, and provides unit-level faithfulness diagnostics. Unless stated otherwise, results use three seeds, which are averaged within each user–domain unit before summarization or inference.

S1. Experimental and Statistical Details

S1.1. Dataset Encoding and First-Occurrence Filter

XPASS-Vis [1] contains 6,526 images, 129 participants, and 87,836 user–image interactions across three visual domains (Tab. S1). The released file encodes aesthetic scores as 0–6 and emotion responses as 0–4. The loader shifts the emotion items to 1–5; leaving the aesthetic score on its 0–6 file coding has no effect on SROCC or PLCC because both are invariant to a constant shift.

Table S1. XPASS-Vis images and rating interactions by domain.

Domain	Images	Interactions
Artwork	2,345	31,543
Fashion	2,082	28,154
Landscape	2,099	28,139
Total	6,526	87,836

Repeated ratings are excluded from all fitting and evaluation. For each repeated user–image pair, we retain the first occurrence in the released file and reserve the second for the test–retest reference. This removes 4,509 rows and leaves 83,327 interactions. The deterministic filter precedes all split-specific processing.

S1.2. Leak-Free Split and Model Selection

The 129 participants are partitioned into 10 fixed groups. Each of five folds uses seven groups for training, one for validation, and two for testing; every participant is a test user in exactly one fold. The exact user counts are shown in Tab. S2. Train, validation, and test participants are disjoint, and images associated with these groups do not cross split boundaries. Stage 1 and the population model are fitted only on training-group images.

Table S2. Fixed participant split used in every experiment.

Fold	Train	Validation	Test	Stage-1 images
0	97	6	26	4,561
1	95	12	22	4,564
2	92	14	23	4,563
3	83	16	30	4,561
4	84	17	28	4,582

Within each user and domain, a deterministic seed (42+user_id) first holds out 50 evaluation images. Support sets of $N \in \{10, 25, 50, 100\}$ are then taken from the remainder, so every rating budget is evaluated on the same images. A candidate personal-head hyperparameter is fitted separately on each validation user's support set and scored by SROCC on that user's disjoint 50-image validation evaluation set. Scores are averaged across validation-user units; one candidate is frozen for the fold–domain–condition–budget combination and applied to all test users. Shared components are selected by validation-group image-level MSE.

Ridge uses 17 penalties from 10^{-2} to 10^{6} (logspace$(-2, 6, 17)$). MLPs have one 128-unit ReLU hidden layer, Adam optimization, MSE loss, no weight decay, and a fixed 500-epoch budget. Their learning rate is selected from $\{10^{-4}, 3\times10^{-4}, \ldots, 3, 10\}$. Early stopping is disabled: at $N = 10$, an internal hold-out would contain only one or two observations.

S1.3. Inference and Holm–Bonferroni Families

SROCC and PLCC are computed on held-out images for each user–domain unit. Domain-specific comparisons use 129 paired participants and two-sided Wilcoxon signed-rank tests. For an Average column, the three domains from one participant are correlated;

therefore the mean uncertainty, confidence interval, and two-sided test use 10,000 participant-cluster bootstrap samples (seed 0), resampling 129 participants and retaining all three domains as a block. Empirical tail probabilities are estimated from those 10,000 samples; a zero tail count is not interpreted as an exact zero and is reported as $p < .001$. Weight-alignment tests are one-sided because the tested alternative is directional and positive.

Holm–Bonferroni correction is applied separately within each family in Tab. S3. SROCC and PLCC are always separate families, and an Average family is never combined with a domain-specific family. Tables not listed there are descriptive and carry no significance markers. The family definitions were fixed before the final correction audit was run; they were not registered in advance.

For traceability, the released statistics CSV also retains 24 exploratory MLP contrasts and 12 exploratory distribution-target contrasts. Their tables below are descriptive: these retained calculations do not create manuscript significance markers or support inferential claims.

S1.4. Reproducibility and Released Artifacts

BLAS operations are pinned to one thread before NumPy is imported. Random projections, permutations, and initializations use fixed seeds; validation ties within a relative tolerance of 10^{-9} are resolved by a deterministic conservative rule. The released uv.lock is the executable reference environment. The final audit used Python 3.13.5, NumPy 2.5.2, pandas 3.0.5, SciPy 1.18.0, scikit-learn 1.9.0, and Matplotlib 3.11.1. Training runs were produced on more than one platform; all reported values are retained as per-unit outputs, and the locked environment regenerates the downstream statistics and presentation artifacts from those outputs.

The emotion, PCA, and identity ridge pipelines are deterministic for a fixed split and support set. Seed averaging therefore affects the stochastic Random and Shuffled controls and the MLP conditions, but does not create three independent versions of each deterministic ridge result. Each participant has one deterministic support-set draw, so the reported uncertainty is conditional on that draw; it does not quantify variation under new support images. Shared fold-level components are likewise treated as fixed by the reported participant-level inference.

The release includes the fixed split files and eleven compact audit-facing CSV files: raw_all_final.csv (seed-level model results), formula_units_final.csv and formula_images_final.csv (formula and image-level diagnostics), stage1_accuracy.csv, statistics/holm_corrected.csv. The set also contains upper_bound_units_final.csv, emotion_importance_units_final.csv, mlp_architecture_units_final.csv, reference_bounds_final.csv, and regularizer_units_final.csv, together with the 387-row test_retest_units.csv used for the within-unit reference. Three small source manifests record the contributing files and their SHA-256 hashes. The licensed raw ratings and large extracted features are not redistributed; their expected schemas and extraction commands are documented in the code package included with this supplementary material.

The frozen per-unit outputs are the numerical source of the release. The locked environment regenerates downstream statistics, tables, and figures, and reproduce.py compares the regenerated Holm table row-for-row with the frozen audit.

S1.5. Reference Quantities

The ground-truth-emotion (GT) reference replaces Stage 1 predictions with each user's reported emotions and fits an unanchored ridge head. Unlike the compared model rows, its penalty is selected per user by RidgeCV on that user's support set. It is therefore an optimistic diagnostic of perfect emotion observation rather than a deployable condition trained under the matched selection rule.

The test–retest reference can be summarized either by pooling all repeated pairs or by first computing a correlation within each user–domain unit. The former is the .684 SROCC reported as a dataset-level reference; the latter is on the same within-user scale as the model scores (Table S4). Because each unit contains a median of only 12 repeated images (range 5–12), the within-unit correlations are themselves noisy and are provided as a scale reference rather than a model comparison.

Table S3. Multiple-comparison families used in the final audit. "Domain" comprises Artwork, Fashion, Landscape, and a separate clustered Average family.

Result block	Family defined by	Reference contrast(s)	Test	Size
Main Table 1	domain × metric	Hybrid vs Population, Direct, Random, PCA	paired / cluster	4
Anchor Table 2	Average × metric	Hybrid vs four mediators, both anchor settings	cluster	8
Backbone Table S2	domain × metric	Hybrid vs Population across five backbones	paired / cluster	5
Efficiency Fig. 4	domain × metric	Hybrid and Direct vs Population at four budgets	paired / cluster	8
Faithfulness tables	domain × metric	formula (6), anchored ablation (3), image (4)	paired / cluster	13
Weight alignment Table 4	domain × coefficient type	alignment greater than zero	one-sided / cluster	1

Table S4. Test–retest correlation under two aggregation rules.

Metric	Aggregation	Art	Fash.	Land.	Avg.
SROCC	Pooled pairs	.672	.688	.695	.684
	Within unit	.577	.593	.584	.585
PLCC	Pooled pairs	.686	.691	.703	.693
	Within unit	.589	.595	.607	.597

Pooled rows contain 4,509 repeated pairs. Within-unit rows average 387 user–domain correlations; the participant-cluster 95% interval for Average SROCC is [.550, .617].

S2. Extended Architecture and Performance Results

S2.1. Visual Backbone Robustness

S2.2. Linear and Nonlinear Heads

The sequential MLP first fits a shared emotion extractor and freezes it before fitting a personal MLP. In the joint variant, the shared extractor is trained with

$$\mathcal{L} = \mathrm{MSE}(\hat{y}) + \lambda\,\mathrm{MSE}(\hat{\mathbf{e}}), \qquad \lambda \in \{0.1, 1, 10\}, \quad \text{(S1)}$$

where the emotion loss is averaged over seven concepts. Task gradients pass through the semantic layer, following the joint concept-bottleneck setup of Koh et al. [2]. Joint training is restricted to the shared stage; it does not fine-tune a high-capacity extractor separately for each user.

S2.3. Detailed Rating-Budget Results

Table S5. Population, Direct, and Hybrid ridge models across visual backbones and domains.

Backbone	Model	Artwork		Fashion		Landscape		Average	
		SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC
CLIP frozen	Population	$.465_{\pm.017}$	$.474_{\pm.016}$	$.326_{\pm.013}$	$.347_{\pm.013}$	$.457_{\pm.015}$	$.459_{\pm.014}$	$.416_{\pm.011}$	$.427_{\pm.011}$
	Direct	$\mathbf{.524_{\pm.014}}$	$\mathbf{.533_{\pm.014}}$	$\mathbf{.365_{\pm.013}}$	$\mathbf{.382_{\pm.013}}$	$.478_{\pm.015}$	$\mathbf{.489_{\pm.014}}$	$\mathbf{.456_{\pm.011}}$	$\mathbf{.468_{\pm.010}}$
	Hybrid	$.486^{\dagger}_{\pm.017}$	$.493^{\dagger}_{\pm.016}$	$.338_{\pm.014}$	$.356^{\dagger}_{\pm.013}$	$\mathbf{.481^{\dagger}_{\pm.014}}$	$.486^{\dagger}_{\pm.014}$	$.435^{\dagger}_{\pm.011}$	$.445^{\dagger}_{\pm.011}$
CLIP-ft (score)	Population	$.470_{\pm.017}$	$.477_{\pm.016}$	$.332_{\pm.013}$	$.347_{\pm.013}$	$.452_{\pm.014}$	$.455_{\pm.014}$	$.418_{\pm.011}$	$.426_{\pm.010}$
	Direct	$\mathbf{.539_{\pm.013}}$	$\mathbf{.542_{\pm.013}}$	$\mathbf{.355_{\pm.013}}$	$\mathbf{.375_{\pm.013}}$	$\mathbf{.482_{\pm.015}}$	$\mathbf{.487_{\pm.014}}$	$\mathbf{.459_{\pm.010}}$	$\mathbf{.468_{\pm.010}}$
	Hybrid	$.488^{\dagger}_{\pm.016}$	$.493_{\pm.016}$	$.340_{\pm.013}$	$.353_{\pm.013}$	$.472^{\dagger}_{\pm.014}$	$.477^{\dagger}_{\pm.014}$	$.434^{\dagger}_{\pm.011}$	$.441^{\dagger}_{\pm.010}$
CLIP-ft (emotion)	Population	$.468_{\pm.016}$	$.473_{\pm.016}$	$.315_{\pm.014}$	$.329_{\pm.013}$	$.435_{\pm.015}$	$.438_{\pm.014}$	$.406_{\pm.011}$	$.413_{\pm.010}$
	Direct	$\mathbf{.525_{\pm.013}}$	$\mathbf{.531_{\pm.013}}$	$\mathbf{.349_{\pm.015}}$	$\mathbf{.362_{\pm.014}}$	$\mathbf{.473_{\pm.015}}$	$\mathbf{.479_{\pm.015}}$	$\mathbf{.449_{\pm.011}}$	$\mathbf{.457_{\pm.011}}$
	Hybrid	$.486^{\dagger}_{\pm.015}$	$.492^{\dagger}_{\pm.015}$	$.328_{\pm.015}$	$.341_{\pm.015}$	$.456^{\dagger}_{\pm.015}$	$.463^{\dagger}_{\pm.015}$	$.423^{\dagger}_{\pm.011}$	$.432^{\dagger}_{\pm.011}$
Qwen3-VL 4B	Population	$.476_{\pm.017}$	$.486_{\pm.016}$	$.363_{\pm.013}$	$.374_{\pm.013}$	$.469_{\pm.014}$	$.472_{\pm.014}$	$.436_{\pm.011}$	$.444_{\pm.011}$
	Direct	$\mathbf{.532_{\pm.013}}$	$\mathbf{.539_{\pm.013}}$	$\mathbf{.382_{\pm.014}}$	$\mathbf{.394_{\pm.013}}$	$\mathbf{.496_{\pm.014}}$	$\mathbf{.501_{\pm.014}}$	$\mathbf{.470_{\pm.011}}$	$\mathbf{.478_{\pm.010}}$
	Hybrid	$.488_{\pm.016}$	$.495_{\pm.016}$	$.365_{\pm.015}$	$.377_{\pm.014}$	$.492^{\dagger}_{\pm.014}$	$.499^{\dagger}_{\pm.014}$	$.448^{\dagger}_{\pm.012}$	$.457^{\dagger}_{\pm.011}$
Qwen3-VL 8B	Population	$.477_{\pm.017}$	$.487_{\pm.017}$	$.372_{\pm.013}$	$.384_{\pm.013}$	$.469_{\pm.015}$	$.475_{\pm.014}$	$.439_{\pm.011}$	$.449_{\pm.011}$
	Direct	$\mathbf{.534_{\pm.014}}$	$\mathbf{.542_{\pm.014}}$	$\mathbf{.398_{\pm.013}}$	$\mathbf{.412_{\pm.013}}$	$\mathbf{.501_{\pm.015}}$	$\mathbf{.510_{\pm.015}}$	$\mathbf{.478_{\pm.011}}$	$\mathbf{.488_{\pm.010}}$
	Hybrid	$.496^{\dagger}_{\pm.016}$	$.504_{\pm.016}$	$.386_{\pm.014}$	$.398^{\dagger}_{\pm.013}$	$.496^{\dagger}_{\pm.015}$	$.504^{\dagger}_{\pm.015}$	$.459^{\dagger}_{\pm.011}$	$.469^{\dagger}_{\pm.011}$

Values are mean ± SEM. Domain columns use 129 participants; Average uses the participant-cluster bootstrap over the same 129 participants.
† marks Hybrid above Population after Holm correction within the corresponding domain–metric family across five backbones. Direct is descriptive and is not included in that family. The best predictive value in each cell block is shown in bold.

Table S6. Same-width Hybrid and PCA mediators across backbones at $N = 100$. Values are Average SROCC.

Backbone	Population-anchored		Unanchored	
	Hybrid	PCA	Hybrid	PCA
CLIP frozen	.435	.445	.419	.377
CLIP-ft (score)	.434	.446	.427	.425
CLIP-ft (emotion)	.423	.441	.419	.421
Qwen3-VL 4B	.448	.454	.436	.362
Qwen3-VL 8B	.459	.458	.446	.382

This descriptive scope analysis adds no Holm contrasts. The anchored Hybrid–PCA near-equality on Qwen3-VL 8B does not extend to the three CLIP variants. Without anchoring, Hybrid is higher on three backbones and differs by at most .003 on the other two.

Table S7. Ridge and sequential MLP heads for each mediator. Qwen3-VL 8B, population-anchored, $N = 100$.

Mediator	Head	Artwork		Fashion		Landscape		Average	
		SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC
Population	Ridge	$.477_{\pm.017}$	$.487_{\pm.017}$	$.372_{\pm.013}$	$.384_{\pm.013}$	$.469_{\pm.015}$	$.475_{\pm.014}$	$.439_{\pm.011}$	$.449_{\pm.011}$
	MLP	$.427_{\pm.016}$	$.438_{\pm.016}$	$.303_{\pm.011}$	$.316_{\pm.012}$	$.401_{\pm.013}$	$.406_{\pm.013}$	$.377_{\pm.010}$	$.386_{\pm.010}$
Direct	Ridge	$.534_{\pm.014}$	$.542_{\pm.014}$	$.398_{\pm.013}$	$.412_{\pm.013}$	$.501_{\pm.015}$	$.510_{\pm.015}$	$.478_{\pm.011}$	$.488_{\pm.010}$
	MLP	$.356_{\pm.014}$	$.354_{\pm.014}$	$.234_{\pm.012}$	$.238_{\pm.012}$	$.336_{\pm.015}$	$.339_{\pm.015}$	$.309_{\pm.010}$	$.310_{\pm.010}$
Random	Ridge	$.495_{\pm.015}$	$.504_{\pm.015}$	$.373_{\pm.013}$	$.384_{\pm.012}$	$.473_{\pm.014}$	$.480_{\pm.014}$	$.447_{\pm.010}$	$.456_{\pm.010}$
	MLP	$.479_{\pm.013}$	$.484_{\pm.014}$	$.328_{\pm.012}$	$.339_{\pm.012}$	$.431_{\pm.014}$	$.431_{\pm.014}$	$.412_{\pm.010}$	$.418_{\pm.010}$
PCA	Ridge	$.510_{\pm.015}$	$.519_{\pm.015}$	$.377_{\pm.014}$	$.388_{\pm.013}$	$.487_{\pm.015}$	$.495_{\pm.015}$	$.458_{\pm.011}$	$.467_{\pm.011}$
	MLP	$.495_{\pm.015}$	$.504_{\pm.015}$	$.350_{\pm.013}$	$.359_{\pm.013}$	$.466_{\pm.016}$	$.469_{\pm.016}$	$.437_{\pm.011}$	$.444_{\pm.011}$
Hybrid (ours)	Ridge	$.496_{\pm.016}$	$.504_{\pm.016}$	$.386_{\pm.014}$	$.398_{\pm.013}$	$.496_{\pm.015}$	$.504_{\pm.015}$	$.459_{\pm.011}$	$.469_{\pm.011}$
	MLP	$.459_{\pm.015}$	$.468_{\pm.016}$	$.315_{\pm.013}$	$.325_{\pm.012}$	$.437_{\pm.015}$	$.441_{\pm.014}$	$.404_{\pm.011}$	$.411_{\pm.010}$

Values are mean ± SEM after averaging seeds within units. Average SEM uses the participant-cluster bootstrap. This table is descriptive and contains no hypothesis tests.

Table S8. Sequential and joint MLP training for the Hybrid mediator. Qwen3-VL 8B, population-anchored, $N = 100$.

Stage-1 → Stage-2	Artwork		Fashion		Landscape		Average	
	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC
Ridge → Ridge (main)	$\mathbf{.496_{\pm.016}}$	$\mathbf{.504_{\pm.016}}$	$\mathbf{.386_{\pm.014}}$	$\mathbf{.398_{\pm.013}}$	$\mathbf{.496_{\pm.015}}$	$\mathbf{.504_{\pm.015}}$	$\mathbf{.459_{\pm.011}}$	$\mathbf{.469_{\pm.011}}$
MLP → MLP, joint $\lambda = 0.1$	$.430_{\pm.016}$	$.437_{\pm.016}$	$.278_{\pm.012}$	$.289_{\pm.012}$	$.392_{\pm.015}$	$.395_{\pm.015}$	$.367_{\pm.010}$	$.374_{\pm.010}$
MLP → MLP, joint $\lambda = 1$	$.450_{\pm.015}$	$.456_{\pm.016}$	$.301_{\pm.012}$	$.315_{\pm.012}$	$.413_{\pm.015}$	$.415_{\pm.014}$	$.388_{\pm.011}$	$.395_{\pm.010}$
MLP → MLP, joint $\lambda = 10$	$.460_{\pm.015}$	$.467_{\pm.015}$	$.310_{\pm.013}$	$.323_{\pm.012}$	$.436_{\pm.015}$	$.439_{\pm.014}$	$.402_{\pm.011}$	$.410_{\pm.011}$
MLP → MLP, sequential	$.459_{\pm.015}$	$.468_{\pm.016}$	$.315_{\pm.013}$	$.325_{\pm.012}$	$.437_{\pm.015}$	$.441_{\pm.014}$	$.404_{\pm.011}$	$.411_{\pm.010}$

Values are mean ± SEM after averaging seeds within units. Average SEM uses the participant-cluster bootstrap. The evaluated MLP variants have lower mean SROCC and PLCC than Ridge → Ridge in every domain. The table is descriptive and contains no hypothesis tests.

Table S9. Descriptive regularizer sensitivity for the personal head. Frozen CLIP, unanchored models, $N = 100$, and three seeds.

Head	Population		Direct		Hybrid	
	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC
Ridge	$.416_{\pm.011}$	$.427_{\pm.011}$	$.400_{\pm.012}$	$.410_{\pm.012}$	$.419_{\pm.011}$	$.429_{\pm.011}$
Lasso	$.412_{\pm.011}$	$.422_{\pm.010}$	$.330_{\pm.012}$	$.340_{\pm.012}$	$.417_{\pm.011}$	$.427_{\pm.011}$
Elastic net	$.412_{\pm.011}$	$.422_{\pm.010}$	$.344_{\pm.012}$	$.354_{\pm.012}$	$.419_{\pm.011}$	$.428_{\pm.011}$

Values are mean ± SEM after averaging the three seeds within each user–domain unit. SEM uses the participant-cluster bootstrap over 129 participants. This table is descriptive and has no significance markers.

Sparse penalties produced a constant prediction in two Lasso–Hybrid units and one Elastic-Net–Hybrid unit, so their correlations are undefined and excluded from the corresponding mean and bootstrap (valid $n = 385$ and 386, respectively). Every other cell contains all 387 units.

All heads were rerun together under the current split, support-set, and validation-selection protocol. Hybrid varies by at most .002 SROCC across the three regularizers, whereas sparse penalties reduce Direct by .055–.070. This supports Ridge as the common head used in the primary experiments.

Table S10. Population, Direct, and Hybrid performance across personal-rating budgets with Qwen3-VL 8B and the population anchor.

N	Model	Artwork		Fashion		Landscape		Average	
		SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC
10	Population	$.477_{\pm.017}$	$.487_{\pm.017}$	$\mathbf{.372_{\pm.013}}$	$\mathbf{.384_{\pm.013}}$	$.469_{\pm.015}$	$.475_{\pm.014}$	$\mathbf{.439_{\pm.011}}$	$\mathbf{.449_{\pm.011}}$
	Direct	$\mathbf{.486_{\pm.016}}$	$\mathbf{.498_{\pm.016}}$	$.368_{\pm.013}$	$.381_{\pm.012}$	$.458_{\pm.015}$	$.464_{\pm.015}$	$.437_{\pm.011}$	$.448_{\pm.011}$
	Hybrid	$.481_{\pm.017}$	$.490_{\pm.016}$	$.347_{\pm.015}$	$.360_{\pm.015}$	$\mathbf{.472_{\pm.015}}$	$\mathbf{.479_{\pm.015}}$	$.433_{\pm.012}$	$.443_{\pm.011}$
25	Population	$.477_{\pm.017}$	$.487_{\pm.017}$	$.372_{\pm.013}$	$.384_{\pm.013}$	$.469_{\pm.015}$	$.475_{\pm.014}$	$.439_{\pm.011}$	$.449_{\pm.011}$
	Direct	$\mathbf{.494_{\pm.016}}$	$\mathbf{.504_{\pm.016}}$	$\mathbf{.376_{\pm.013}}$	$\mathbf{.389_{\pm.013}}$	$\mathbf{.481_{\pm.014}}$	$\mathbf{.489_{\pm.014}}$	$\mathbf{.450_{\pm.011}}$	$\mathbf{.460_{\pm.011}}$
	Hybrid	$.478_{\pm.017}$	$.487_{\pm.016}$	$.368_{\pm.014}$	$.380_{\pm.014}$	$.478_{\pm.015}$	$.486_{\pm.015}$	$.442_{\pm.012}$	$.451_{\pm.011}$
50	Population	$.477_{\pm.017}$	$.487_{\pm.017}$	$.372_{\pm.013}$	$.384_{\pm.013}$	$.469_{\pm.015}$	$.475_{\pm.014}$	$.439_{\pm.011}$	$.449_{\pm.011}$
	Direct	$\mathbf{.516_{\pm.014}}$	$\mathbf{.524_{\pm.014}}$	$\mathbf{.382_{\pm.013}}$	$\mathbf{.393_{\pm.012}}$	$\mathbf{.493_{\pm.014}}$	$\mathbf{.503_{\pm.014}}$	$\mathbf{.464_{\pm.011}}$	$\mathbf{.473_{\pm.010}}$
	Hybrid	$.488_{\pm.016}$	$.496_{\pm.016}$	$.374_{\pm.014}$	$.383_{\pm.013}$	$.489_{\pm.015}$	$.499_{\pm.015}$	$.450_{\pm.011}$	$.459_{\pm.011}$
100	Population	$.477_{\pm.017}$	$.487_{\pm.017}$	$.372_{\pm.013}$	$.384_{\pm.013}$	$.469_{\pm.015}$	$.475_{\pm.014}$	$.439_{\pm.011}$	$.449_{\pm.011}$
	Direct	$\mathbf{.534_{\pm.014}}$	$\mathbf{.542_{\pm.014}}$	$\mathbf{.398_{\pm.013}}$	$\mathbf{.412_{\pm.013}}$	$\mathbf{.501_{\pm.015}}$	$\mathbf{.510_{\pm.015}}$	$\mathbf{.478_{\pm.011}}$	$\mathbf{.488_{\pm.010}}$
	Hybrid	$.496_{\pm.016}$	$.504_{\pm.016}$	$.386_{\pm.014}$	$.398_{\pm.013}$	$.496_{\pm.015}$	$.504_{\pm.015}$	$.459_{\pm.011}$	$.469_{\pm.011}$

Values are mean ± SEM after averaging the three seeds within each unit. Domain SEM uses 129 participants; Average SEM uses the participant- cluster bootstrap. This numerical breakdown is descriptive; the inferential tests attached to Fig. 4 use the families in Tab. S3.

S3. Mediator-Design Analysis

S3.1. Stage-1 Emotion Extraction

Recovery varies substantially by concept: the pooled SROCC is highest for Amused, Distasteful, and Impressed (.597–.607) and lowest for Sad (.445). This table measures visual recoverability, not downstream importance; the next analysis separates associations in users' reported emotions from the weights available to the deployable predicted-emotion model.

S3.2. Learned Emotion Weights

Amused has the largest GT magnitude (.444), whereas Impressed ranks first under predicted emotions (.287). These rank changes show why concept predictability and personal association should be reported separately. The coefficients are descriptive properties of fitted linear heads, not causal effects of inducing an emotion.

S3.3. Distribution-Valued Emotion Targets

The main model predicts the seven population-mean emotion ratings. We also tested a 14-dimensional target containing each mean and across-rater standard deviation, and a 35-dimensional target containing five rating-bin frequencies for each emotion. These variants ask whether disagreement information omitted by the mean is useful downstream.

Table S11. Stage-1 emotion extraction accuracy by visual domain.

Concept	Artwork		Fashion		Landscape		Average	
	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC
Amused	$.684_{\pm.036}$	$.696_{\pm.033}$	$.450_{\pm.045}$	$.463_{\pm.048}$	$\mathbf{.687_{\pm.023}}$	$\mathbf{.687_{\pm.023}}$	$\mathbf{.607_{\pm.035}}$	$.615_{\pm.035}$
Distasteful	$.682_{\pm.023}$	$.686_{\pm.018}$	$\mathbf{.562_{\pm.031}}$	$\mathbf{.601_{\pm.028}}$	$.569_{\pm.022}$	$.583_{\pm.024}$	$.604_{\pm.020}$	$\mathbf{.623_{\pm.017}}$
Impressed	$\mathbf{.694_{\pm.029}}$	$.704_{\pm.029}$	$.449_{\pm.033}$	$.472_{\pm.032}$	$.649_{\pm.014}$	$.665_{\pm.016}$	$.597_{\pm.032}$	$.614_{\pm.031}$
Intellectual	$.487_{\pm.052}$	$.503_{\pm.053}$	$.371_{\pm.041}$	$.384_{\pm.047}$	$.563_{\pm.039}$	$.597_{\pm.031}$	$.474_{\pm.032}$	$.495_{\pm.033}$
Motivated	$.545_{\pm.039}$	$.587_{\pm.044}$	$.346_{\pm.048}$	$.362_{\pm.044}$	$.537_{\pm.033}$	$.550_{\pm.035}$	$.476_{\pm.033}$	$.500_{\pm.034}$
Nostalgic	$.683_{\pm.022}$	$\mathbf{.717_{\pm.026}}$	$.378_{\pm.022}$	$.395_{\pm.025}$	$.617_{\pm.027}$	$.673_{\pm.019}$	$.559_{\pm.037}$	$.595_{\pm.040}$
Sad	$.556_{\pm.036}$	$.573_{\pm.036}$	$.317_{\pm.029}$	$.375_{\pm.023}$	$.460_{\pm.037}$	$.512_{\pm.028}$	$.445_{\pm.032}$	$.487_{\pm.027}$

Values are mean ± SEM across the five fold-level estimates in each domain; Average summarizes the 15 fold–domain estimates. This shared-stage analysis is descriptive and has no user-level hypothesis test.

Table S12. Mean absolute standardized emotion weights (± SEM), percentage of positive coefficients, and within-setting importance rank.

Emotion	Artwork	Fashion	Landscape	Average	Rank
Ground-Truth (GT)					
Amused	$.451_{\pm.024}$ (98.4%)	$.419_{\pm.022}$ (96.9%)	$.463_{\pm.021}$ (100.0%)	$.444_{\pm.019}$ (98.4%)	1
Impressed	$.314_{\pm.018}$ (97.7%)	$.224_{\pm.014}$ (91.5%)	$.259_{\pm.015}$ (97.7%)	$.266_{\pm.014}$ (95.6%)	2
Distasteful	$.235_{\pm.015}$ (8.5%)	$.270_{\pm.019}$ (7.8%)	$.195_{\pm.014}$ (8.5%)	$.233_{\pm.013}$ (8.3%)	3
Intellectual	$.178_{\pm.010}$ (82.2%)	$.136_{\pm.009}$ (76.0%)	$.133_{\pm.008}$ (82.2%)	$.149_{\pm.007}$ (80.1%)	4
Motivated	$.119_{\pm.008}$ (75.2%)	$.142_{\pm.010}$ (80.6%)	$.154_{\pm.011}$ (89.1%)	$.139_{\pm.007}$ (81.7%)	5
Nostalgic	$.129_{\pm.008}$ (83.7%)	$.085_{\pm.006}$ (59.7%)	$.091_{\pm.006}$ (70.5%)	$.102_{\pm.004}$ (71.3%)	6
Sad	$.101_{\pm.008}$ (66.7%)	$.098_{\pm.008}$ (32.6%)	$.087_{\pm.006}$ (42.6%)	$.095_{\pm.004}$ (47.3%)	7
Predicted (Pred)					
Impressed	$.459_{\pm.024}$ (92.2%)	$.156_{\pm.008}$ (88.4%)	$.244_{\pm.012}$ (96.9%)	$.287_{\pm.009}$ (92.5%)	1
Distasteful	$.229_{\pm.015}$ (24.8%)	$.218_{\pm.012}$ (10.1%)	$.174_{\pm.011}$ (14.0%)	$.207_{\pm.009}$ (16.3%)	2
Amused	$.206_{\pm.015}$ (82.2%)	$.133_{\pm.011}$ (82.9%)	$.207_{\pm.011}$ (96.1%)	$.182_{\pm.008}$ (87.1%)	3
Motivated	$.187_{\pm.016}$ (47.3%)	$.144_{\pm.013}$ (79.1%)	$.112_{\pm.007}$ (70.5%)	$.148_{\pm.008}$ (65.6%)	4
Sad	$.163_{\pm.012}$ (54.3%)	$.136_{\pm.012}$ (55.0%)	$.119_{\pm.009}$ (55.0%)	$.139_{\pm.007}$ (54.8%)	5
Intellectual	$.151_{\pm.010}$ (48.1%)	$.116_{\pm.009}$ (56.6%)	$.129_{\pm.009}$ (40.3%)	$.132_{\pm.006}$ (48.3%)	6
Nostalgic	$.192_{\pm.013}$ (51.2%)	$.096_{\pm.008}$ (46.5%)	$.108_{\pm.007}$ (46.5%)	$.132_{\pm.006}$ (48.1%)	7

Domain columns contain 129 user units. Average magnitude SEM uses a participant-cluster bootstrap retaining all three domains; percentages use a 10^{-9} tolerance for numerical zeros and are averaged across domains.

GT fits a diagnostic ridge head directly to the user's same-session reported emotions, with its penalty selected per user by RidgeCV on that user's support set. Pred reports the anchored model's total emotion-space weight $\mathbf{w}_{\text{tot}}$. The blocks are descriptive interpretations, not a controlled predictive comparison between identical training rules.

Table S13. Distribution-valued Stage-1 targets with frozen CLIP, a population anchor, and a ridge head. Widths are 7 (means), 14 (means and standard deviations), and 35 (seven five-bin histograms).

N	Population	Direct	Hybrid 7-d	Hybrid 14-d	Hybrid 35-d
10	$.416_{\pm.011}$	$.420_{\pm.011}$	$.414_{\pm.011}$	$.419_{\pm.011}$	$.413_{\pm.011}$
25	$.416_{\pm.011}$	$.431_{\pm.011}$	$.415_{\pm.011}$	$.419_{\pm.011}$	$.425_{\pm.011}$
50	$.416_{\pm.011}$	$.440_{\pm.011}$	$.423_{\pm.011}$	$.428_{\pm.011}$	$.435_{\pm.011}$
100	$.416_{\pm.011}$	$.456_{\pm.011}$	$.435_{\pm.011}$	$.434_{\pm.011}$	$.442_{\pm.011}$

Values are pooled mean SROCC ± participant-cluster bootstrap SEM over 129 participants. The table is descriptive. At $N = 100$, the 14-dimensional and 7-dimensional targets are nearly identical (.434 and .435). The histogram target is highest from $N = 25$ onward, but uses five times the width of the named bottleneck. We therefore retain the seven-dimensional mean target for the primary study.

S4. Faithfulness Diagnostics

All diagnostics operate on the stored formulas and evaluation-image records; they do not refit any model. Formula swaps use five fixed-seed peers sampled without replacement within the same fold and domain; their uncertainty is therefore conditional on that fixed peer assignment. Image swaps use every one of the 49 nonzero cyclic shifts of a unit's 50 evaluation images. The "no-image" condition supplies the user's support-set mean emotion vector, so it retains no image-specific emotion information. This is a baseline emotion vector rather than setting the seven raw emotion ratings to zero.

The diagnostics use two weight conventions for distinct estimands. Concept ablation ranks named concepts by the reported total emotion-space weight $\mathbf{w}_{\text{tot}} = \mathbf{w}_{\text{pop}} + \boldsymbol{\delta}_u$. Image swap reconstructs the executed anchored predictor, $\hat{y}_i = \hat{y}_{\text{pop}}(x_i) + \mathbf{z}_i^\top \boldsymbol{\delta}_u + b_u$, and changes only $\mathbf{z}_i$. Formula swap instead replaces the stored personal coefficient and intercept with five eligible peer formulas; it uses the frozen $K = 5$ protocol.

S4.1. Qualitative Example and Failure Case

Figure S1 makes the numerical formula concrete. The example is the top-ranked candidate produced by the disclosed search in qualitative_triple_search.py; the rule requires an opposite-sign Sad-weight pair with a gap of at least .30, minimizes the pair's maximum prediction error on a shared image, and then shows the largest-error third user on that image. The figure is illustrative rather than an additional hypothesis test; the aggregate diagnostics above carry the statistical claims.

Table S14. Formula-swap diagnostic. "Other user" averages five deterministic peer swaps ($K = 5$) within the same fold and domain.

Setting	Formula	Artwork		Fashion		Landscape		Average	
		SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC
Anchored	Own	$\mathbf{.496}_{\pm.016}$	$\mathbf{.504}_{\pm.016}$	$\mathbf{.386}_{\pm.014}$	$\mathbf{.398}_{\pm.013}$	$\mathbf{.496}_{\pm.015}$	$\mathbf{.504}_{\pm.015}$	$\mathbf{.459}_{\pm.011}$	$\mathbf{.469}_{\pm.011}$
	Population mean	$.476^{\dagger}_{\pm.017}$	$.487^{\dagger}_{\pm.017}$	$.371_{\pm.013}$	$.384^{\dagger}_{\pm.013}$	$.470^{\dagger}_{\pm.015}$	$.475^{\dagger}_{\pm.014}$	$.439^{\dagger}_{\pm.011}$	$.449^{\dagger}_{\pm.011}$
	Other user	$.420^{\dagger,\ddagger}_{\pm.016}$	$.428^{\dagger,\ddagger}_{\pm.015}$	$.338^{\dagger,\ddagger}_{\pm.013}$	$.351^{\dagger,\ddagger}_{\pm.012}$	$.429^{\dagger,\ddagger}_{\pm.013}$	$.430^{\dagger,\ddagger}_{\pm.013}$	$.396^{\dagger,\ddagger}_{\pm.010}$	$.403^{\dagger,\ddagger}_{\pm.010}$
Unanchored	Own	$\mathbf{.490}_{\pm.014}$	$\mathbf{.493}_{\pm.015}$	$\mathbf{.368}_{\pm.014}$	$\mathbf{.381}_{\pm.014}$	$\mathbf{.479}_{\pm.014}$	$\mathbf{.488}_{\pm.014}$	$\mathbf{.446}_{\pm.011}$	$\mathbf{.454}_{\pm.011}$
	Population mean	$.466^{\dagger}_{\pm.016}$	$.474^{\dagger}_{\pm.016}$	$.355_{\pm.013}$	$.366^{\dagger}_{\pm.013}$	$.454^{\dagger}_{\pm.015}$	$.460^{\dagger}_{\pm.014}$	$.425^{\dagger}_{\pm.011}$	$.434^{\dagger}_{\pm.010}$
	Other user	$.402^{\dagger,\ddagger}_{\pm.014}$	$.408^{\dagger,\ddagger}_{\pm.014}$	$.305^{\dagger,\ddagger}_{\pm.012}$	$.319^{\dagger,\ddagger}_{\pm.012}$	$.398^{\dagger,\ddagger}_{\pm.013}$	$.399^{\dagger,\ddagger}_{\pm.013}$	$.368^{\dagger,\ddagger}_{\pm.009}$	$.376^{\dagger,\ddagger}_{\pm.009}$

Domain columns report mean ± SEM over 129 participants; the Average column uses participant-cluster bootstrap SEM over the same 129 participants, retaining their three domains as a block.
† marks a Holm-adjusted two-sided difference from Own; ‡ marks Population mean versus Other user. All contrasts are members of the joint faithfulness family in Tab. S3.

Table S15. Concept-ablation diagnostic under the population anchor. One emotion is held at its support-set mean according to the magnitude of $\mathbf{w}_{\text{tot}} = \mathbf{w}_{\text{pop}} + \boldsymbol{\delta}_u$.

Condition	Artwork		Fashion		Landscape		Average	
	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC
Full	$\mathbf{.496}_{\pm.016}$	$\mathbf{.504}_{\pm.016}$	$\mathbf{.386}_{\pm.014}$	$\mathbf{.398}_{\pm.013}$	$\mathbf{.496}_{\pm.015}$	$\mathbf{.504}_{\pm.015}$	$\mathbf{.459}_{\pm.011}$	$\mathbf{.469}_{\pm.011}$
Top-weight held	$.303^{\dagger}_{\pm.024}$	$.307^{\dagger}_{\pm.023}$	$.338^{\dagger}_{\pm.014}$	$.343^{\dagger}_{\pm.014}$	$.443^{\dagger}_{\pm.017}$	$.452^{\dagger}_{\pm.017}$	$.361^{\dagger}_{\pm.013}$	$.367^{\dagger}_{\pm.012}$
Average-weight held	$.457^{\dagger}_{\pm.016}$	$.466^{\dagger}_{\pm.016}$	$.371^{\dagger}_{\pm.014}$	$.381^{\dagger}_{\pm.013}$	$.478^{\dagger}_{\pm.015}$	$.486^{\dagger}_{\pm.015}$	$.435^{\dagger}_{\pm.011}$	$.444^{\dagger}_{\pm.011}$
Bottom-weight held	$.495_{\pm.016}$	$.503_{\pm.016}$	$.385_{\pm.014}$	$.397_{\pm.014}$	$.496_{\pm.015}$	$.504_{\pm.015}$	$.459_{\pm.011}$	$.468^{\dagger}_{\pm.011}$

Domain columns report mean ± SEM over 129 participants; the Average column uses participant-cluster bootstrap SEM. † marks a Holm-adjusted two-sided difference from Full in the anchored faithfulness family. The Bottom-weight SROCC comparison is not detected ($p = .628$), whereas its PLCC comparison is detected ($p = .036$).
The raw-feature population prediction remains fixed. Thus $\mathbf{w}_{\text{tot}}$ provides formula-level concept attribution rather than a path decomposition of the population branch or a residual-only ablation of $\boldsymbol{\delta}_u$.

Table S16. Image-swap diagnostic. The fitted formula and target image's population prediction are fixed while the emotion input is replaced.

Setting	Emotion input	Artwork		Fashion		Landscape		Average	
		SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC
Anchored	Correct image	$\mathbf{.496}_{\pm.016}$	$\mathbf{.504}_{\pm.016}$	$\mathbf{.386}_{\pm.014}$	$\mathbf{.398}_{\pm.013}$	$\mathbf{.496}_{\pm.015}$	$\mathbf{.504}_{\pm.015}$	$\mathbf{.459}_{\pm.011}$	$\mathbf{.469}_{\pm.011}$
	No-image emotion	$.477^{\dagger}_{\pm.017}$	$.487_{\pm.017}$	$.372_{\pm.013}$	$.384^{\dagger}_{\pm.013}$	$.469^{\dagger}_{\pm.015}$	$.475^{\dagger}_{\pm.014}$	$.439^{\dagger}_{\pm.011}$	$.449^{\dagger}_{\pm.011}$
	Swapped image	$.414^{\dagger,\ddagger}_{\pm.015}$	$.423^{\dagger,\ddagger}_{\pm.015}$	$.319^{\dagger,\ddagger}_{\pm.012}$	$.332^{\dagger,\ddagger}_{\pm.012}$	$.418^{\dagger,\ddagger}_{\pm.013}$	$.423^{\dagger,\ddagger}_{\pm.013}$	$.384^{\dagger,\ddagger}_{\pm.010}$	$.392^{\dagger,\ddagger}_{\pm.010}$
Unanchored	Correct image	$\mathbf{.490}_{\pm.014}$	$\mathbf{.493}_{\pm.015}$	$\mathbf{.368}_{\pm.014}$	$\mathbf{.381}_{\pm.014}$	$\mathbf{.479}_{\pm.014}$	$\mathbf{.488}_{\pm.014}$	$\mathbf{.446}_{\pm.011}$	$\mathbf{.454}_{\pm.011}$
	Swapped image	$-.010^{\dagger}_{\pm.000}$	$-.010^{\dagger}_{\pm.000}$	$-.008^{\dagger}_{\pm.000}$	$-.008^{\dagger}_{\pm.000}$	$-.010^{\dagger}_{\pm.000}$	$-.010^{\dagger}_{\pm.000}$	$-.009^{\dagger}_{\pm.000}$	$-.009^{\dagger}_{\pm.000}$

Domain columns report mean ± SEM over 129 participants; the Average column uses participant-cluster bootstrap SEM. Swapped-image results average all 49 nonzero cyclic shifts of each unit's 50 evaluation images.
† marks a Holm-adjusted two-sided difference from Correct image in the same anchoring condition; ‡ marks No-image emotion versus Swapped image. All contrasts belong to the joint faithfulness family in Tab. S3.
For the unanchored swapped-image row, both SEMs are below .001 and therefore round to .000 at the table's three-decimal precision. The unanchored No-image predictor is constant across a user's evaluation images, so SROCC and PLCC are undefined and that row is omitted.
Correct image versus Swapped image tests image specificity. In the anchored model, No-image emotion is the image-dependent population prediction plus a constant, so its SROCC matches the Population baseline by construction.



Image: Art - 4173

	impressed	intellectual	motivated	amused	nostalgic	sad	distasteful	predicted ŷ	Actual y
Predicted Emotion (ê)	2.705	2.867	2.043	2.719	1.989	**2.035**	1.859		
Personal Weight (w)									
User 106:	0.176	-0.184	0.446	0.150	-0.332	**0.191**	-0.541	3.98	4
User 129:	0.699	0.102	-0.288	0.006	0.095	**-0.230**	-0.030	4.03	4
User 92:	0.734	-0.377	0.581	-0.345	0.011	**0.334**	-0.153	3.38	2

Figure S1. Qualitative personal-formula example and failure case. All three users rated the same artwork (image 4173). The first row shows the Stage-1 emotion vector on its 1–5 scale; these values are standardized before entering the personal head. The displayed coefficients are total emotion-space weights $\mathbf{w}_{\text{tot}}$. The predictions are those of the complete anchored model, including the raw-feature population prediction and personal intercept. Users 106 and 129 have opposite-sign Sad weights ($w_{\text{sad}} = +.191$ and $-.230$) and predictions within .03 of their recorded scores. User 92 has the largest error among the remaining users who rated the image ($y = 2.0$, $\hat{y} = 3.38$) and is shown as a failure case.

Table S17. Alignment between each fitted total emotion-space weight vector and the corresponding user's empirical emotion–rating associations.

Domain	Spearman	Pearson
Artwork	$.418^{\dagger}_{\pm.023}$	$.446^{\dagger}_{\pm.019}$
Fashion	$.542^{\dagger}_{\pm.024}$	$.579^{\dagger}_{\pm.021}$
Landscape	$.615^{\dagger}_{\pm.022}$	$.622^{\dagger}_{\pm.021}$
Average	$.525^{\dagger}_{\pm.015}$	$.549^{\dagger}_{\pm.013}$

For each user–domain unit, the seven entries of $\mathbf{w}_{\text{tot}}$ are correlated with the seven associations between predicted emotion and raw aesthetic rating. Values are mean ± SEM over 129 participants. The Average SEM uses the participant-cluster bootstrap, retaining all three domains per participant.
† marks a positive-alignment result under the directional one-sided test. Each domain–coefficient-type family contains one contrast, as listed in Tab. S3.
Domain intervals use the normal approximation (mean ±1.96 SEM); the Average interval is a participant-cluster bootstrap percentile interval.

S5. Scope of the Evidence

The seven personal coefficients describe how an individual's prediction departs from a population branch fitted on raw visual features; they do not decompose that shared branch. The computational diagnostics provide evidence for owner, weight, and image specificity within the evaluated model. No participant inspected or edited a formula in this study, so usability and successful human correction remain questions for a user study. Comparisons with published ICI and MIR numbers are descriptive only because their features and protocols differ; the content-free Random control also reaches a similar range.

References

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[2] Pang Wei Koh, Thao Nguyen, Yew Siang Tang, Stephen Mussmann, Emma Pierson, Been Kim, and Percy Liang. Concept bottleneck models. In Proceedings of the 37th International Conference on Machine Learning, pages 5338–5348. PMLR, 2020. 3